Flexible Monolithic 3D Integration of RRAM-based CIM, CNT CMOS Circuits and IGZO FET-based Sensor and Display for Augmented Reality Contact Lenses

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***Abstract*—Augmented reality (AR) contact lenses emerge as a promising immersive platform offering seamless, eye-integrated augmented experiences. In this work, for the first time, we present **M3D-FAR**, a prototype **M**onolithic **3D** integration chip featuring **F**lexible **A**ugmented **R**eality contact lenses with three interconnected functional layers on the same flexible substrate: 1st layer of 8×8 Ta_2O_5-based resistive random-access memory (RRAM) array for digital computing-in-memory (CIM), 2nd layer of carbon nanotube (CNT) CMOS circuits for logic and data interface, and 3rd layer of $InGaZnO_x$ field-effect transistors (IGZO-FET) for 8×24 photosensor array and display driver circuits. The structural integrity and proper function of the M3D-FAR chip was validated by structural analysis and electrical measurements. Furthermore, system-level benchmark in a typical image-based interactive task shows that the M3D-FAR architecture could achieve 10.45× speed-up compared to its 2D counterpart and consume 5.39× lower energy than GPU.

I. Introduction

AR contact lenses enable seamless, eye-integrated augmented reality but require biocompatible, flexible chips capable of fast image processing with integrated sensing and display (**Fig. 1a-b**) [1]. The M3D-FAR chip architecture addresses this demand by vertically stacking a RRAM-based CIM layer, a CNT-CMOS logic/interface layer, and an IGZO-FET photosensor/display layer on a flexible polyimide (PI) substrate, connected via ultra-dense inter-layer vias (ILVs) [2-5]. This M3D-FAR architecture delivers image recognition accuracy comparable to its 2D counterpart while reducing the energy consumption by 5.39× and increasing processing speed by 10.45×. Fig. 1c shows the lift-off and packaging of the flexible M3D-FAR chip, while Fig. 1d highlights its key advantages.

II. Design and Fabrication of M3D-FAR Chip

The M3D-FAR chip was fabricated on a flexible PI substrate using a low-temperature (<300 °C) monolithic 3D process. The first CIM layer consists of Ta_2O_5-based RRAM array with a material stack of Ti/Pd/Ta_2O_5/Ta (5/25/10/40 nm), passivated by 25 nm Al_2O_3/10 nm HfO_2. The second CNT-CMOS layer consists of 40 nm Pt as the gate, 3 nm Al_2O_3/7 nm HfO_2 by ALD as the gate dielectric, solution-processed high-density CNT film as the channel, 40 nm Pd as the source/drain contacts, 10 nm HfO_2/10 nm AlN by ALD as the electrostatic doping for NMOS and thermally oxidized 3 nm Y as the passivation layer for PMOS. The third IGZO-FET layer uses 10 nm Ti/15 nm Pd as the back gate, 20 nm Al_2O_3 by ALD as the gate oxide, 20 nm sputtered IGZO as the channel, and 20 nm Ti/40 nm Pd as the source/drain contacts. After fabrication, the chip was packaged with biocompatible PHEMA for AR contact lenses. **Fig. 2** shows the fabrication process flow, optical images of the stacked device layers, as well as optical and TEM images of each layer.

III. Electrical Measurement Results

The M3D-FAR chip heterogeneously integrates three layers of devices, including CNT-FETs, IGZO-FETs and RRAMs. **Fig. 3–5** present the device characterizations. RRAMs exhibit DC I–V switching curves, robust resistance states under bending, large HRS/LRS window, stable retention, and correct CIM cell operation. The CNT-FETs show a large on/off ratio across 1,000 devices, and enable D Flip-Flop logic (DFF), NAND gates, and 170 ring oscillators (ROs) to operate correctly at V_{DD} = 5 V, confirming high-speed logic functionality. Meanwhile, the flexible IGZO-FETs maintain stable transfer characteristics under bending with I_{ON}/I_{OFF} >10^8 for 60 devices, pronounced photo-response to illumination and 5 s pulses, and uniform material composition confirmed by EDS mapping.

Fig. 6 shows the optical images and electrical measurements of the CNT-CMOS Amplifier (AMP) and trans-impedance Amplifier (TIA), demonstrating an open-loop gain exceeding 1,000 and high linearity respectively. **Fig. 7a-c** present the performance of the IGZO-FET driver cell, achieving an ON/OFF ratio above 10^5. **Fig. 7d-e** confirm the successful fabrication and correct operation of ALU at 10 kHz.

Fig. 8 illustrates the workflow of M3D-FAR chip architecture, where the IGZO-FET sensor photocurrents are routed via high-density ILVs to the CNT-CMOS layer for logic conversion and CIM-based computation, and the results are then sent to the IGZO-FET display drivers to generate pixel currents for interactive output. **Fig. 9** shows the key circuit modules, LED brightness control, and programmable photocurrents from the 8×24 photosensor array.

IV. System-Level Benchmark

Fig. 10a depicts the M3D-FAR chip implementing real-time SR and segmentation via lightweight Swin-IR and Mobile-SAM. **Fig. 10b** benchmarks the system showing 10.45× faster updates, 5.39× lower energy, and 20× smaller area compared to its 2D counterpart. The comparison in **Table 1** highlights that this work reports the first flexible M3D architecture that monolithically integrates three different layers with five functionalities [3-10].

V. Conclusions

To sum up, we have successfully demonstrated a M3D-FAR chip for flexible AR contact lenses by monolithically integrating RRAM-based CIM array, CNT-CMOS based logic circuits and data interface, and IGZO-FET based photosensor array and display driver circuits. Electrical measurements validate its proper functionality, and a system-level benchmark shows that M3D-FAR could achieve 5.39× lower energy consumption and 10.45× faster inference speed than its 2D counterpart.

Acknowledgements

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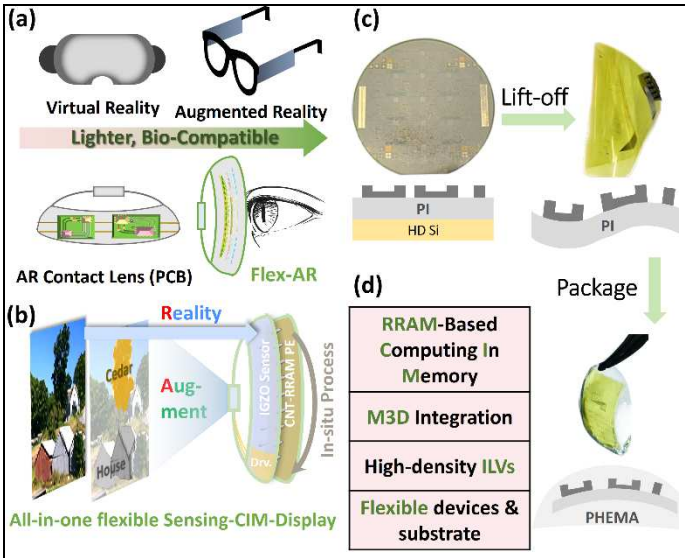


Fig.1 (a) Evolution from VR to flexible AR contact lenses. (b) Design of the M3D-FAR chip architecture. (c)Lift-off and packaging of flexible M3D-FAR chip. (d) Advantages.

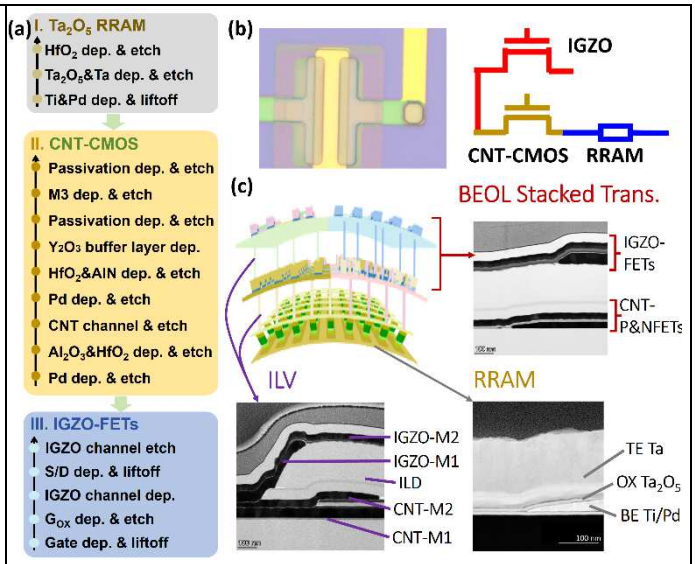


Fig.2 (a) Fabrication process flow of the flexible M3D-FAR chip. (b)Photo of three functional layers. (c) Cross-sectional TEM images.

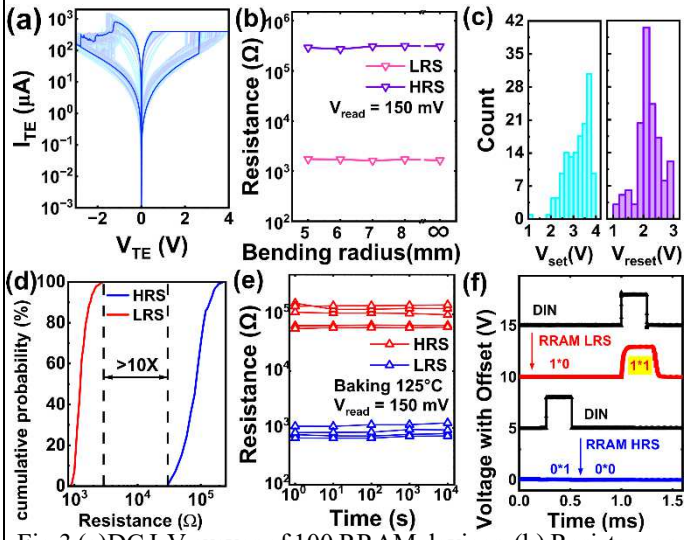


Fig.3 (a)DC I-V curves of 100 RRAM devices. (b) Resistances measured at different bending radii. (c) Distribution of the set and reset voltages. (d)The cumulative probability of resistance states. (e) Retention test. (f) Waveforms of a typical CIM cell.

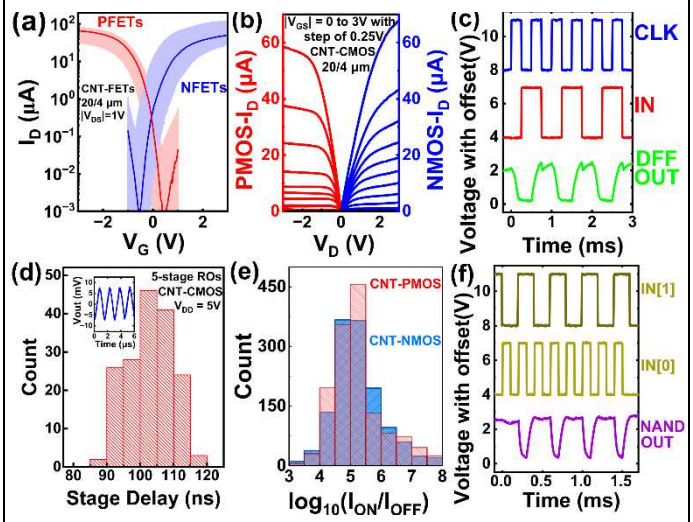


Fig. 4 (a)I_{DS}-V_G transfer curves of 1000 CMOS CNTFETs. (b) I_{DS}-V_D output curves. (c) Waveforms of DFF. (d) Waveforms and stage delay of 170 ROs at V_{DD}=5V. (e) I_{ON}/I_{OFF} histogram. (f) Waveforms of NAND at 10 kHz.

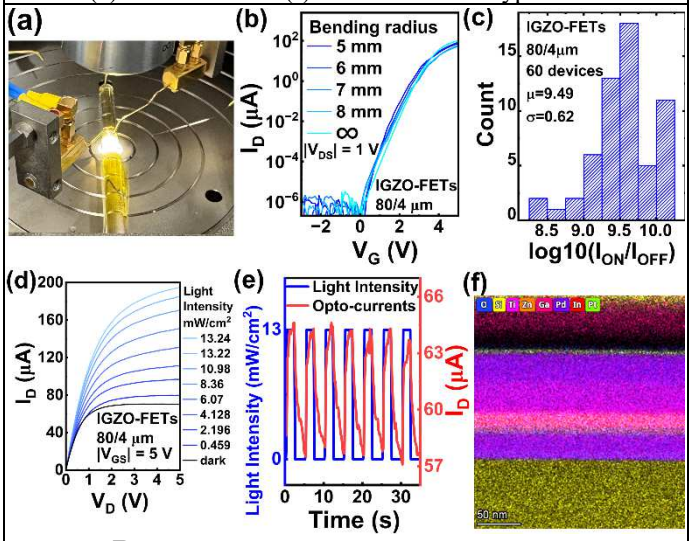


Fig.5 (a)Bending tests of IGZO-FET (b)I_{DS}-V_G curves of at different bending radii. (c)I_{ON}/I_{OFF} histogram of 60 flexible IGZO-FETs. (d)Photoelectrical response (e)Current of illuminated with 5s light pulses. (f) EDS mapping picture.

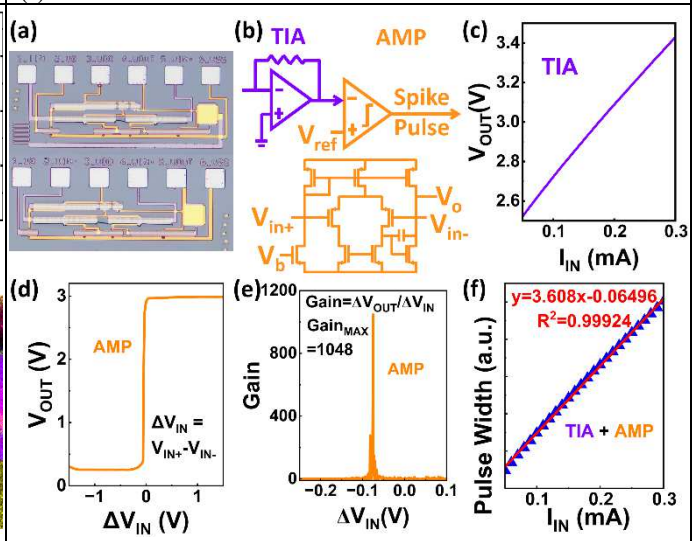


Fig.6 (a) Optical images of TIA and AMP. (b) Schematic of TIA and AMP. (c)Output curve of CNT CMOS TIA. (d)Output curve of CNT CMOS AMP. (e)Output gain of AMP (f)Linear performance of CNT CMOS TIA and AMP.

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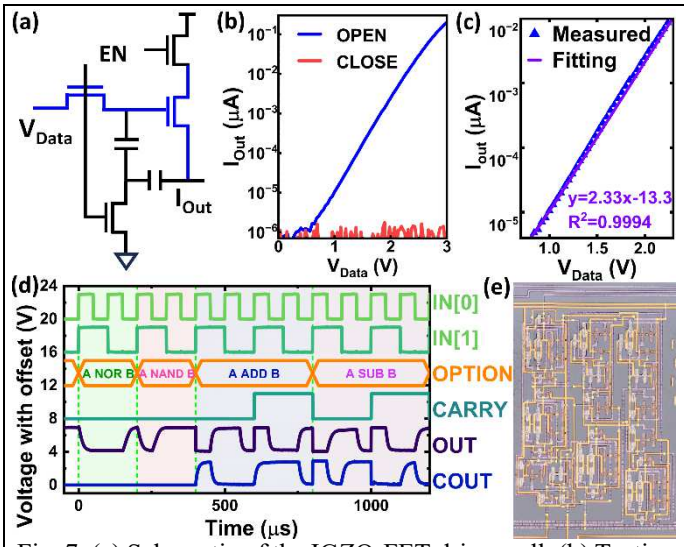


Fig. 7. (a) Schematic of the IGZO-FET driver cell. (b) Testing of IGZO-FET driver cell. (c) Linear performance of IGZO-FET driver cell. (d) Waveforms of CNT CMOS ALU with 4 functions at 10 kHz. (e) Photo of CNT-CMOS ALU.

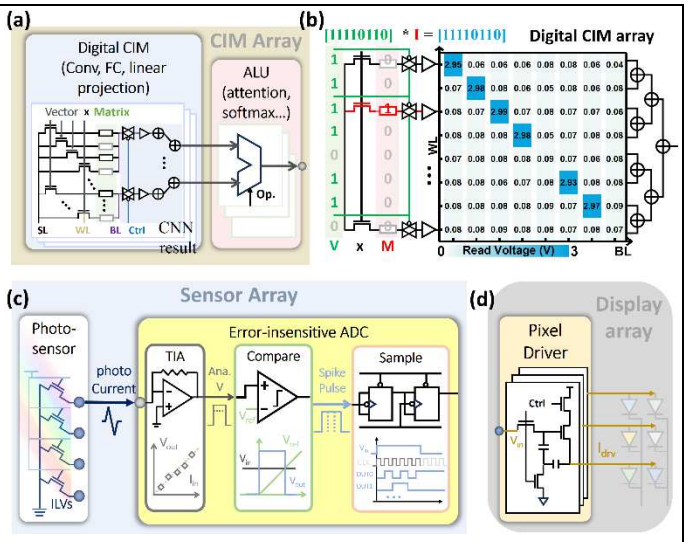


Fig. 8 (a) Workflow of the CIM array. (b) Diagram and MVM computation process using the CNT-RRAM digital CIM array. (c) Workflow of the sensor array. (d) Workflow of the display array.

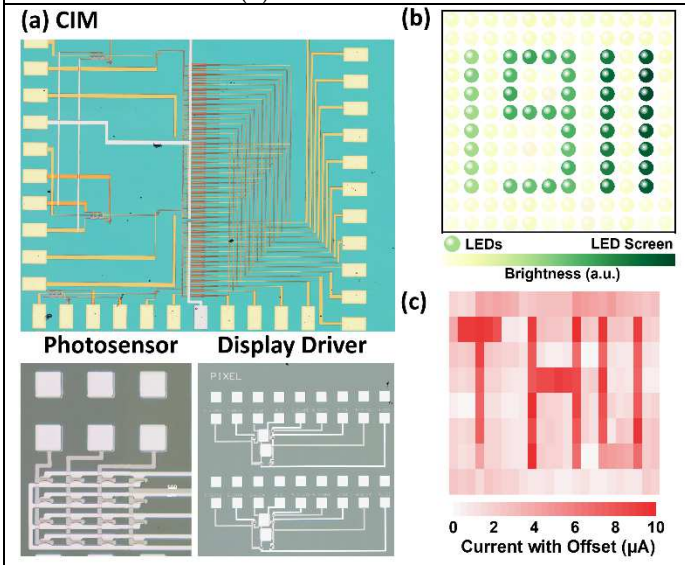
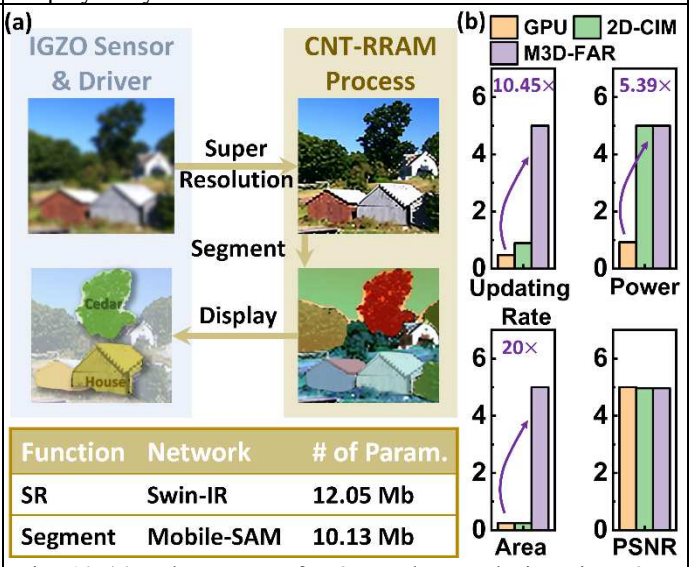


Fig. 9 (a) Optical images of the key circuit modules. (b) LED brightness driven by the display. (c) Photocurrents of an 8×24 photosensor array.



Function	Network	# of Param.
SR	Swin-IR	12.05 Mb
Segment	Mobile-SAM	10.13 Mb

Fig. 10 (a) Advantages of M3D and CIM designs in M3D-FAR. (b) Performance benchmark of M3D-FAR against its 2D counterpart and GPU.

Ref.	BEOL Devices		Integ-ration	Flexi-bility	On-chip Functionalities				
	Trans.	Mem.			Sense	Inter-face	CIM	Logic	Dis-play
[3]	CNT	RRAM	M3D	/	YES	/	YES	YES	/
[4]	IGZO	RRAM	M3D	/	YES	/	YES	/	/
[5]	IGZO	/	/	YES	/	/	/	YES	/
[6]	MoS_2	/	M3D	/	/	/	/	/	YES
[7]	/	RRAM	M3D		YES	YES	YES	/	/
[8]	IGZO	/	/	YES	YES	YES	/	/	/
[9]	ITO	/	M3D	YES	/	/	/	/	YES
[10]	ITO	/	M3D	/	YES	/	/	/	YES
This Work	CNT IGZO	RRAM	M3D	YES	YES	YES	YES	YES	YES

Table. 1 Comparison with prior works of M3D integration.

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